

Lab 02 - First Verilog

In this lab, you've learned how to do an initial and simple design in Verilog to learn the Vivado tooling and process involved in RTL/FPGA design.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

This week's lab taught me how to set up a Verilog project in Vivado using pre-existing files and how to alter a file to assign a switch to an LED with Verilog. I then learned how to simulate what a file will do to a Basys board with Vivado's built in simulator. Once the simulation matched expected output, I then learned how to Generate a bitstream and set up a physical board for the code in Vivado to run on. I was then able to check that my code was doing what was expected by flipping a switch on the Basys board and watching the corresponding light turn on.

Lab Questions

1 - Describe the stages of building a Verilog project in Vivado.

The first stage is to create a project with source files top.v, test.v, and constraints file constraints.xd. When adding them, you set them up as simulation only for test.v and simulation and synthesis for top.v. Once the project is created the next stage is to run a simulation to see what Vivado expects the top.v file to do on a Basys board. It is good practice to simulate before the next stages, where Vivado will first run synthesis to create a netlist, then implementation where it maps the design synthesis generated onto the specific chip being used, after that it will place and route the circuit onto the board and generate the bitstream using this information which makes the circuit actually come to life on our board.

2 - What is the value in looking at the elaborated design schematic?

The value of looking at the schematic is you can see if your code is generating a schematic that looks like the diagram that you are trying to replicate.

3 - Why should we simulate our designs frequently? What does the simulation do?

The simulation is going to tell you what the circuit is going to output based on the state of the inputs. It's also good to simulate as it is faster than compiling and downloading, it's good for quickly checking if the changes you make are producing expected output.

Code Submission

https://github.com/TaogAnderson/week_2_first_verilog.git